FASTR Workshop - Test Country

January 15-17, 2026 | Test Country

Workshop Facilitator

FASTR
FREQUENT ASSESSMENTS & SYSTEMS TOOLS FOR RESILIENCE

Workshop Agenda - Day 1

Time	Session	Speaker
8:30 AM - 9:00 AM	**Registration**	
9:00 AM - 9:15 AM	**Welcome & Opening Remarks**	
9:15 AM - 9:45 AM	**Participant Introductions**	
9:45 AM - 10:00 AM	**Workshop Objectives**	
10:00 AM - 10:20 AM	**Country Overview**	
10:20 AM - 10:35 AM	**Health Priorities**	
10:35 AM - 10:50 AM	*Tea Break*	
10:50 AM - 12:05 PM	**Introduction to the FASTR Approach**	
12:05 PM - 12:30 PM	**Data Extraction**	
12:30 PM - 1:30 PM	*Lunch*	
1:30 PM - 2:35 PM	**Data Extraction**	
2:35 PM - 3:30 PM	**Group Work & Discussion**	
3:30 PM - 3:45 PM	*Afternoon Tea*	
3:45 PM - 4:45 PM	**Group Work & Discussion**	
4:45 PM - 5:00 PM	**Day Wrap-up & Q&A**	

Workshop Agenda - Day 2

Time	Session	Speaker
9:00 AM - 9:15 AM	**Recap & Questions**	
9:15 AM - 10:30 AM	**Data Quality Assessment**	
10:30 AM - 10:45 AM	*Tea Break*	
10:45 AM - 11:45 AM	**Data Quality Adjustment**	
11:45 AM - 12:30 PM	**Hands-on Practice: DQ Adjust**	
12:30 PM - 1:30 PM	*Lunch*	
1:30 PM - 3:30 PM	**Group Work & Discussion**	
3:30 PM - 3:45 PM	*Afternoon Tea*	
3:45 PM - 4:45 PM	**Group Work & Discussion**	
4:45 PM - 5:00 PM	**Day Wrap-up & Q&A**	

Workshop Agenda - Day 3

Time	Session	Speaker
9:00 AM - 9:15 AM	**Recap & Questions**	
9:15 AM - 10:15 AM	**Data Analysis**	
10:15 AM - 10:30 AM	**Data Analysis**	
10:30 AM - 10:45 AM	*Tea Break*	
10:45 AM - 11:00 AM	**Data Analysis**	
11:00 AM - 12:30 PM	**Hands-on Practice: Analysis**	
12:30 PM - 1:30 PM	*Lunch*	
1:30 PM - 3:30 PM	**Group Work & Discussion**	
3:30 PM - 3:45 PM	*Afternoon Tea*	
3:45 PM - 4:45 PM	**Group Work & Discussion**	
4:45 PM - 5:05 PM	**Next Steps & Action Planning**	
5:05 PM - 5:20 PM	**Day Wrap-up & Q&A**	

Workshop Objectives

By the end of this workshop, participants will be able to:

- EPI, MCH
- English
- 10
- 4

Country Health System Overview

Test Country

Health System Structure

Level	Description
National	[Ministry of Health]
Regional	[X provinces/regions]
District	[X districts]
Facility	500 health facilities

Reporting to DHIS2: 400 (80)

Population

Group	Estimate
Total population	1000000
Women of reproductive age	200000
Children under 5	150000
Expected pregnancies/year	[X per year]
Expected live births/year	[X per year]

Data Sources

Routine data:

- DHIS2 (reporting rate: 80)

Survey data:

- DHIS2 Monthly

Health Priorities

Focus areas for Test Country

- Maternal health services (ANC, skilled birth attendance)
- Child immunization coverage
- Family planning services
- Data quality improvement

☕ Tea Break

15 minutes

Back at 10:50 AM

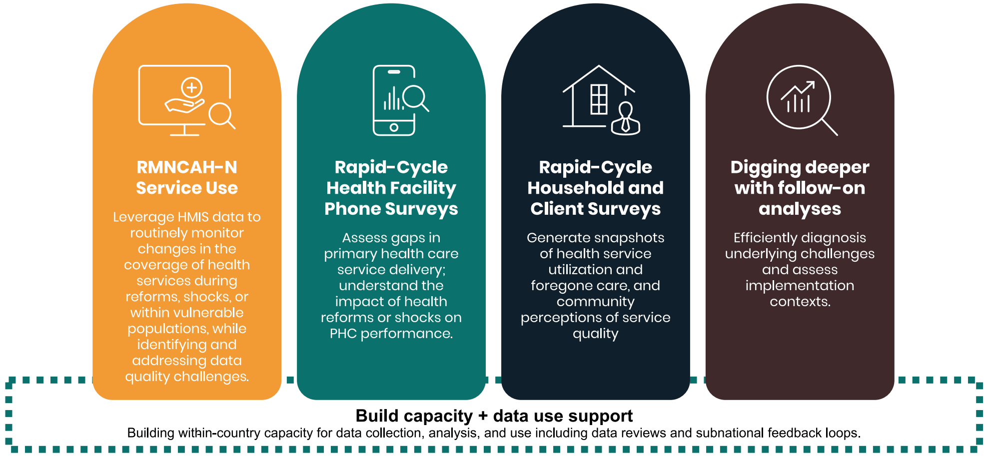
Introduction to FASTR

The Global Financing Facility (GFF) supports country-led efforts to strengthen the use of timely data for decision-making, with the goal of improving primary healthcare (PHC) performance and RMNCAH-N outcomes.

Frequent Assessments and Health System Tools for Resilience (FASTR) is the GFF's rapid-cycle analytics framework for monitoring health system performance using high-frequency data.

FASTR brings together four complementary technical approaches:

1. Routine HMIS data analysis
2. Health facility phone surveys
3. High-frequency household phone surveys
4. Follow-on, problem-driven analyses



What FASTR does with routine HMIS data

FASTR works directly with Ministries of Health to transform routine HMIS data into actionable evidence for policy and program management.

Using facility-level data, the approach focuses on three core analytic functions:

Assess data quality

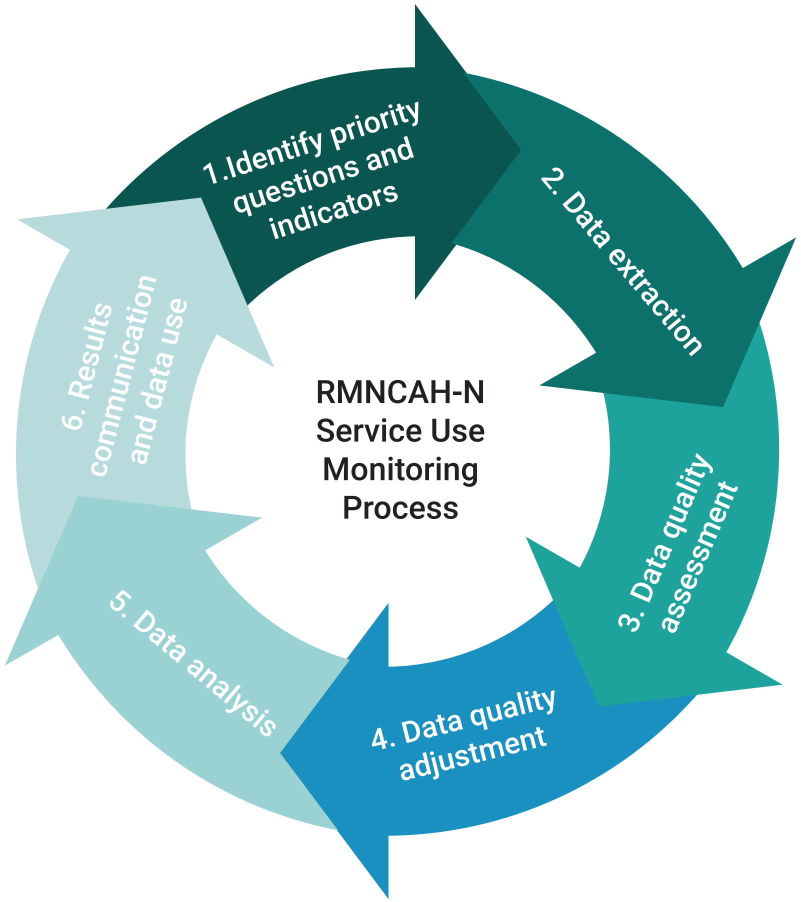
Identify key issues related to completeness, outliers, and internal consistency.

Adjust for data quality limitations

Apply transparent, indicator-specific methods to improve the reliability of trend analysis.

Analyze service use and coverage trends

Track changes in priority RMNCAH-N services and compare progress against country priorities and benchmarks.



Why rapid-cycle analytics?

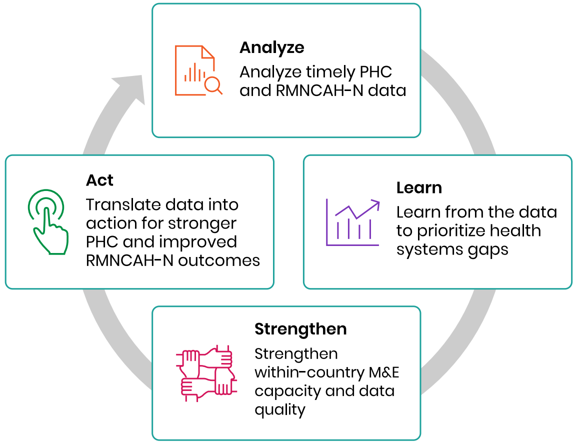
Routine health information systems are a critical source of data, but they are often underused due to concerns about data quality and long delays between data collection and analysis. Traditional household and facility surveys, while essential, are resource-intensive and infrequent.

FASTR's rapid-cycle analytics address this gap by providing:

- Timely insights aligned with country decision cycles
- Continuous learning rather than one-off assessments
- Direct feedback loops between data, analysis, and action

During the COVID-19 pandemic, this approach was applied in over 20 countries to monitor disruptions to essential RMNCAH-N services and inform response and recovery planning.

FASTR's rapid-cycle analytics approach: Analyze, learn, strengthen, act



Focus of the analysis

Core indicators

FASTR prioritizes a core set of RMNCAH-N indicators that:

- Represent key service delivery contacts across the continuum of care
- Have relatively high reporting completeness and volumes
- Serve as proxies for broader service delivery performance

Outpatient consultations are included as a proxy for overall health service use. The indicator set can be expanded to reflect country-specific priorities.

Core data quality metrics

Analysis is anchored in a standardized set of data quality metrics, including:

- Reporting completeness
- Extreme value (outlier) detection
- Consistency across related indicators

These metrics are summarized into an overall data quality score to support interpretation and comparison across areas.

FASTR approach to routine data analysis

The FASTR approach follows a three-step workflow:

1. Assess data quality

Identify issues related to completeness, outliers, and internal consistency at national and subnational levels.

2. Adjust for data quality limitations

Apply transparent, indicator-specific corrections to improve the reliability of trend analysis.

3. Analyze service delivery

Quantify changes in priority service volumes and compare coverage trends against country targets.

This enables continuous, subnational monitoring while data quality is systematically improved.

Why extract data from DHIS2?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

The FASTR methodology includes specific approaches to:

- Identify and adjust for outliers
- Adjust for incomplete reporting
- Apply consistent data quality metrics

These adjustments require processing that cannot be done within DHIS2's native analytics.

Tools for data extraction

Content to be developed

This section will cover:

- DHIS2 data export options
- API-based extraction methods
- Data transformation requirements
- Quality checks on extracted data

🍽️ Lunch Break

60 minutes

Back at 1:30 PM

Why extract data from DHIS2?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

The FASTR methodology includes specific approaches to:

- Identify and adjust for outliers
- Adjust for incomplete reporting
- Apply consistent data quality metrics

These adjustments require processing that cannot be done within DHIS2's native analytics.

Tools for data extraction

Content to be developed

This section will cover:

- DHIS2 data export options
- API-based extraction methods
- Data transformation requirements
- Quality checks on extracted data

☕ Afternoon Tea

15 minutes

Back at 3:45 PM

See You Tomorrow!

Day 1 Complete

We covered today:

- Introduction to the FASTR Approach
- Data Extraction

Tomorrow: Day 2

We resume at **9:00 AM**

Coming up: Data Quality Assessment, Data Quality Adjustment

Day 2: Recap & Questions

Yesterday we covered:

- Introduction to the FASTR Approach
- Data Extraction

Questions & Discussion

- Any questions from yesterday's sessions?
- Points that need clarification?
- Insights from the exercises?

Data quality assessment

Understanding the reliability of routine health data

Why talk about data quality?

The challenge: Health facilities report data every month, but sometimes:

- Numbers seem too high or too low
- Facilities forget to report
- Related numbers don't match up

The impact: Bad data leads to bad decisions

- We might think services are improving when they're not
- We might miss real problems in certain areas
- Resources might go to the wrong places

FASTR's solution: Check data quality systematically, fix what we can, and be transparent about limitations

Three simple questions about data quality

1. Are facilities reporting regularly?

- Completeness: Did we get reports from facilities this month?

2. Are the numbers reasonable?

- Outliers: Are there any suspiciously high values?

3. Do related numbers make sense together?

- Consistency: Do related services show expected patterns?

These three questions help us understand if we can trust the data for decision-making.

Question 1: Are facilities reporting?

Completeness: Did we get reports?

What we're checking:
Each month, are facilities sending in their reports?

Example:

- District has 20 health centers
- In March, only 15 sent ANC data
- **Completeness = 75%** (15 out of 20 reported)

Why it matters:

- If many facilities don't report, we're missing part of the picture
- Trends might look like services dropped, when really facilities just didn't report

What's good completeness?

It depends on your health system:

- 90%+ is excellent
- 80-90% is good
- Below 80% means we're missing a lot of information

Important: Even 100% completeness doesn't mean we have the full picture - some services might happen outside facilities or some facilities might not be in the reporting system.

What to look for: Is completeness improving over time? Which areas have low completeness?

Completeness: FASTR output

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above
80% to 89%
Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

Question 2: Are numbers reasonable?

Outliers: Spotting suspicious numbers

What we're checking:
Are there any values that seem way too high compared to what that facility normally reports?

Real example:

- Health Center A normally reports 20-25 deliveries per month
- In March, they reported 450 deliveries
- **This is likely a data entry error** (maybe they typed an extra digit, or reported cumulative instead of monthly)

Why it matters:

- One extreme value can make it look like there was a huge service increase
- Skews totals and trends for the whole district or province

How we spot outliers

Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator.

A value is flagged as an outlier if it meets EITHER of two criteria:

1. A value greater than 10 times the Median Absolute Deviation (MAD) from the monthly median value for the indicator, OR
2. A value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%

AND for which the count is greater than 100.

Outlier example

Health Center B - Malaria tests:

Month	Tests Reported	Normal?
January	245	Normal
February	267	Normal
March	**2,890**	**Outlier**
April	256	Normal

What happened? Probably someone entered "2890" instead of "289" (extra zero)

Impact if we don't fix it: March would show a huge "spike" in malaria that didn't really happen.

Outliers: FASTR output

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Question 3: Do related numbers match up?

Consistency: Do related services make sense together?

What we're checking:

Health services are related - certain patterns are expected.

Example 1 - ANC visits:

- More women should get their **1st** ANC visit (ANC1)
- Fewer should complete all **4** visits (ANC4)
- We expect: ANC1 >= ANC4

Example 2 - Vaccinations:

- More babies should get their **1st** Penta dose (Penta1)
- Fewer should complete all **3** doses (Penta3)
- We expect: Penta1 >= Penta3

If these relationships are backwards, something's wrong with the data.

Why check consistency at district level?

Patients move between facilities:

- Woman might get ANC1 at Health Center A
- But deliver at District Hospital B
- If we only look at each facility separately, numbers might not match

Solution: Check consistency at district level

- Add up all ANC1 visits in the district
- Add up all ANC4 visits in the district
- Compare the totals

This accounts for patients visiting different facilities for different services.

Consistency example

District X - ANC Services:

Indicator	District Total	Expected Relationship
ANC1	5,200 visits	Should be higher
ANC4	4,100 visits	Should be lower

This passes the consistency check - more women started ANC (5,200) than completed 4 visits (4,100).

If it was reversed (more ANC4 than ANC1), we'd know there's a data quality problem.

Consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

90% or above
80% to 89%
Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughy the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Putting it all together: Overall data quality

Overall quality score

For each facility and month, we combine all three checks:

Complete: Did the facility report?
No outliers: Are the numbers reasonable?
Consistent: Do related numbers make sense?

Binary DQA Score:

- dqa_score = 1 if all consistency pairs pass
- dqa_score = 0 if any consistency pair fails

DQA Mean: Average of completeness-outlier score and consistency score

This score helps us:

- Decide which data to use for analysis
- Identify facilities that need support
- Track if data quality is improving over time

Overall DQA score: FASTR output

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above
70% to 79%
Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: FASTR output

Mean DQA score

Average data quality score across facility-months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

80% or above
70% to 79%
Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

☕ Tea Break

15 minutes

Back at 10:45 AM

Approach to data quality adjustment

The Data Quality Adjustment module (Module 2 in the FASTR analytics platform) systematically corrects two common problems in routine health facility data:

1. **Outliers** - extreme values caused by reporting errors or data entry mistakes
2. **Missing data** - from incomplete reporting

Rather than simply deleting problematic data, this module replaces questionable values with statistically sound estimates based on each facility's own historical patterns.

Four adjustment scenarios

The module produces four parallel versions of the data:

Scenario	Description
None	Original data, no adjustments
Outliers only	Only outlier corrections applied
Completeness only	Only missing data filled in
Both	Both types of corrections applied

This allows analysts to understand how sensitive their results are to different data quality assumptions.

Adjustment for outliers

For each value flagged as an outlier, the module calculates what the value "should have been" based on that facility's historical pattern.

Methods used (in order of preference):

1. Centered 6-month rolling average (3 months before + 3 months after)
2. Forward 6-month rolling average
3. Backward 6-month rolling average
4. Same month from the previous year (for seasonal indicators)
5. Facility-specific historical mean (fallback)

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausibility of the data.

Adjustment for completeness

For months where data is missing or marked as incomplete, the module imputes (fills in) values using the same rolling average approach.

This ensures that temporary reporting gaps don't create artificial drops to zero in the data.

Completeness adjustment: FASTR output

Deviance Due to Incompleteness

Percent change in volume due to completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.7%	3.1%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.6%	8.3%
Bombali District	2.6%	3.3%	4.1%	10.4%	12.9%	7.4%	5.9%	5.6%	13.0%	5.6%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	24.3%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.3%	0.3%	12.8%	7.5%	1.8%	4.7%
Kailahun District	0.5%	1.9%	1.3%	13.8%	11.9%	0.5%	0.4%	0.4%	8.3%	8.1%	2.8%	4.4%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	9.6%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	8.3%	6.3%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	0.1%	0.2%	0.2%	5.6%	4.5%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.6%	1.5%	2.1%	2.3%	41.1%	23.9%	13.1%	14.9%
Kono District	0.4%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	6.8%	7.3%	1.2%	6.5%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	13.9%	7.4%	2.8%	5.3%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.9%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.3%	0.9%
Tonkolili District	1.2%	3.0%	4.6%	6.5%	2.7%	1.5%	1.8%	1.7%	9.6%	8.2%	2.5%	3.7%
Western Area Rural District	0.7%	0.8%	1.7%	1.7%	2.0%	0.6%	2.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.6%	6.4%	19.9%	15.5%	4.5%	6.4%

3% or above
1% to 2%
Below 1%

Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. The deviance is the difference in volume after imputing incomplete data. High levels of deviance can affect the plausibility of the data.

Heatmap showing percent change in service volumes due to missing data imputation

🍽️ Lunch Break

60 minutes

Back at 1:30 PM

☕ Afternoon Tea

15 minutes

Back at 3:45 PM

See You Tomorrow!

Day 2 Complete

We covered today:

- Data Quality Assessment
- Data Quality Adjustment

Tomorrow: Day 3

We resume at **9:00 AM**

Coming up: Data Analysis

Day 3: Recap & Questions

Yesterday we covered:

- Data Quality Assessment
- Data Quality Adjustment

Questions & Discussion

- Any questions from yesterday's sessions?
- Points that need clarification?
- Insights from the exercises?

Service utilization analysis

The Service Utilization module (Module 3 in the FASTR analytics platform) analyzes health service delivery patterns to detect and quantify disruptions in service volumes over time.

Key capabilities:

- Identifies when health services deviate significantly from expected patterns
- Measures magnitude of disruptions at national, provincial, and district levels
- Distinguishes normal fluctuations from genuine disruptions requiring investigation

Two-stage analysis process

Part 1: Control chart analysis

- Model expected patterns using historical trends and seasonality
- Detect significant deviations from expected volumes
- Flag disrupted periods

Part 2: Disruption quantification

- Use panel regression to estimate service volume changes
- Calculate shortfalls and surpluses in absolute numbers

Surplus and disruption analyses

The module detects multiple types of service disruptions:

Disruption Type	Description
Sharp disruptions	Single months with extreme deviations
Sustained drops	Gradual declines over several months
Sustained dips	Periods consistently below expected levels
Sustained rises	Periods consistently above expected levels
Missing data patterns	Gaps in reporting that may signal problems

Quantifying impact

Disruption analysis quantifies shortfalls and surpluses by comparing:

- **Predicted volumes** (what would have happened without disruption)
- **Actual volumes** (what was observed)

Results are reported in absolute numbers and percentages at each geographic level.

Service Coverage Estimates

This module estimates health service coverage by answering: **"What percentage of the target population received this health service?"**

Three data sources integrated:

1. Adjusted health service volumes from HMIS
2. Population projections from United Nations
3. Household survey data from MICS/DHS

Two-Part Process

Part 1: Denominator Calculation

- Calculate target populations using multiple methods (HMIS-based and population-based)
- Compare against survey benchmarks
- Automatically select best denominator for each indicator

Part 2: Coverage Estimation

- Override automatic selections based on programmatic knowledge
- Project survey estimates forward using HMIS trends
- Generate final coverage estimates

Service utilization analysis

The Service Utilization module (Module 3 in the FASTR analytics platform) analyzes health service delivery patterns to detect and quantify disruptions in service volumes over time.

Key capabilities:

- Identifies when health services deviate significantly from expected patterns
- Measures magnitude of disruptions at national, provincial, and district levels
- Distinguishes normal fluctuations from genuine disruptions requiring investigation

Two-stage analysis process

Part 1: Control chart analysis

- Model expected patterns using historical trends and seasonality
- Detect significant deviations from expected volumes
- Flag disrupted periods

Part 2: Disruption quantification

- Use panel regression to estimate service volume changes
- Calculate shortfalls and surpluses in absolute numbers

Surplus and disruption analyses

The module detects multiple types of service disruptions:

Disruption Type	Description
Sharp disruptions	Single months with extreme deviations
Sustained drops	Gradual declines over several months
Sustained dips	Periods consistently below expected levels
Sustained rises	Periods consistently above expected levels
Missing data patterns	Gaps in reporting that may signal problems

Quantifying impact

Disruption analysis quantifies shortfalls and surpluses by comparing:

- **Predicted volumes** (what would have happened without disruption)
- **Actual volumes** (what was observed)

Results are reported in absolute numbers and percentages at each geographic level.

Service Coverage Estimates

This module estimates health service coverage by answering: **"What percentage of the target population received this health service?"**

Three data sources integrated:

1. Adjusted health service volumes from HMIS
2. Population projections from United Nations
3. Household survey data from MICS/DHS

Two-Part Process

Part 1: Denominator Calculation

- Calculate target populations using multiple methods (HMIS-based and population-based)
- Compare against survey benchmarks
- Automatically select best denominator for each indicator

Part 2: Coverage Estimation

- Override automatic selections based on programmatic knowledge
- Project survey estimates forward using HMIS trends
- Generate final coverage estimates

☕ Tea Break

15 minutes

Back at 10:45 AM

Service utilization analysis

The Service Utilization module (Module 3 in the FASTR analytics platform) analyzes health service delivery patterns to detect and quantify disruptions in service volumes over time.

Key capabilities:

- Identifies when health services deviate significantly from expected patterns
- Measures magnitude of disruptions at national, provincial, and district levels
- Distinguishes normal fluctuations from genuine disruptions requiring investigation

Two-stage analysis process

Part 1: Control chart analysis

- Model expected patterns using historical trends and seasonality
- Detect significant deviations from expected volumes
- Flag disrupted periods

Part 2: Disruption quantification

- Use panel regression to estimate service volume changes
- Calculate shortfalls and surpluses in absolute numbers

Surplus and disruption analyses

The module detects multiple types of service disruptions:

Disruption Type	Description
Sharp disruptions	Single months with extreme deviations
Sustained drops	Gradual declines over several months
Sustained dips	Periods consistently below expected levels
Sustained rises	Periods consistently above expected levels
Missing data patterns	Gaps in reporting that may signal problems

Quantifying impact

Disruption analysis quantifies shortfalls and surpluses by comparing:

- **Predicted volumes** (what would have happened without disruption)
- **Actual volumes** (what was observed)

Results are reported in absolute numbers and percentages at each geographic level.

Service Coverage Estimates

This module estimates health service coverage by answering: **"What percentage of the target population received this health service?"**

Three data sources integrated:

1. Adjusted health service volumes from HMIS
2. Population projections from United Nations
3. Household survey data from MICS/DHS

Two-Part Process

Part 1: Denominator Calculation

- Calculate target populations using multiple methods (HMIS-based and population-based)
- Compare against survey benchmarks
- Automatically select best denominator for each indicator

Part 2: Coverage Estimation

- Override automatic selections based on programmatic knowledge
- Project survey estimates forward using HMIS trends
- Generate final coverage estimates

🍽️ Lunch Break

60 minutes

Back at 1:30 PM

☕ Afternoon Tea

15 minutes

Back at 3:45 PM

Next Steps

Action items

- Apply FASTR methods to your country data
- Share findings with stakeholders
- Establish regular data review processes
- Identify areas for data quality improvement

Thank You!

Questions & Discussion

Contact Information

FASTR Team

📧 Email: fastr@worldbank.org
🌐 Website: https://fastr.org